

[EMBEDDED C PROGRAMMING ASSIGNMENT]

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**COURSE :- BTECH (ELECTRICAL
ENGINEERING)**

**SUBJECT :- PROGRAMMING FUNDAMENTALS
FOR ELECTRONICS ENGINEERS**

ADMISSION NUMBER :- 25SECE1050025

1. Write a Program to set(enable) the 5th bit of a it register .

#CODE

```
#include <stdio.h>

int main() {
    int reg;
    printf("Enter register value: ");
    scanf("%d", &reg);
    reg = reg | (1 << 4);
    printf("Value after setting 5th bit: %d\n", reg);

    return 0;
}
```

#OUTPUT#

Output

Clear

Enter register value: 4
Value after setting 5th bit: 20

=== Code Execution Successful ===

2. Write a program to clear the 2nd bit of a 16-bit register .

```
#code
```

```
#include <stdio.h>
```

```
int main() {
```

```
    unsigned short reg;
```

```
    printf("Enter register value: ");
```

```
    scanf("%hu", &reg);
```

```
    reg = reg & ~(1 << 1);
```

```
    printf("Value after clearing 2nd bit: %hu\n", reg);
```

```
    return 0;
```

```
}
```

#OUTPUT`#

Output

Clear

Enter register value: 7

Value after clearing 2nd bit: 5

=== Code Execution Successful ===

3. Write a program to toggle the 7th bit of an 8 bit register.

```
#code
```

```
#include <stdio.h>
```

```
int main() {
```

```
    unsigned char reg;
```

```
    printf("Enter register value: ");
```

```
    scanf("%hhu", &reg);
```

```
    reg = reg ^ (1 << 6);
```

```
    printf("Value after toggling 7th bit: %hhu\n", reg);
```

```
    return 0;
```

```
}
```

#OUTPUT#

Output

Clear

```
Enter register value: 10
Value after toggling 7th bit: 74

=== Code Execution Successful ===
```

4. Write a program to check whether a given bit position is set or not .(4th bit)

#code

#include <stdio.h>

int main() {

int num;

printf("Enter a number: ");

scanf("%d", &num);

if (num & (1 << 3)) {

printf("4th bit is SET\n");

} else {

printf("4th bit is NOT SET\n");

}

return 0;

}

#OUTPUT#

Output

Clear

Enter a number: 10

4th bit is SET

=== Code Execution Successful ===

5. Write a program to count the number of set bits(1s) in a number.

```
#code
```

```
#include <stdio.h>
```

```
int main() {
```

```
    int num, count = 0;
```

```
    printf("Enter a number: ");
```

```
    scanf("%d", &num);
```

```
    while (num > 0) {
```

```
        if (num & 1) {
```

```
            count++;
```

```
        }
```

```
        num = num >> 1;
```

```
    }
```

```
    printf("Number of set bits = %d\n", count);
```

```
    return 0;
```

```
}
```

#OUTPUT#

Output

Clear

```
Enter a number: 4  
Number of set bits = 1
```

```
=== Code Execution Successful ===
```

6. Write a program to perform left shift and right shift operations on a number and display result .

#code

#include <stdio.h>

int main() {

int num, leftShift, rightShift;

printf("Enter a number: ");

scanf("%d", &num);

leftShift = num << 1;

rightShift = num >> 1;

printf("After Left Shift by 1: %d\n", leftShift);

printf("After Right Shift by 1: %d\n", rightShift);

return 0;

}

#OUTPUT#

```
Output Clear  
Enter a number: 8  
After Left Shift by 1: 16  
After Right Shift by 1: 4  
  
=== Code Execution Successful ===
```

7. Write a program to swap two numbers using bitwise XOR.

#code

#include <stdio.h>

int main() {

int a, b;

printf("Enter first number: ");

scanf("%d", &a);

printf("Enter second number: ");

scanf("%d", &b);

printf("\nBefore swapping:\n");

printf("a = %d\nb = %d\n", a, b);

a = a ^ b;

b = a ^ b;

a = a ^ b;

printf("\nAfter swapping:\n");

printf("a = %d\nb = %d\n", a, b);

return 0;

}

#OUTPUT#

Output

Clear

```
Enter first number: 10  
Enter second number: 23
```

```
Before swapping:
```

```
a = 10  
b = 23
```

```
After swapping:
```

```
a = 23  
b = 10
```

```
=== Code Execution Successful ===
```

8. Write a program to clear all bits except a specific bit position.

```
#code
```

```
#include <stdio.h>
```

```
int main() {
```

```
    int num, pos, result;
```

```
    printf("Enter a number: ");
```

```
    scanf("%d", &num);
```

```
    printf("Enter bit position to keep (0-based index): ");
```

```
    scanf("%d", &pos);
```

```
    result = num & (1 << pos);
```

```
    printf("Result after clearing all bits except position %d =  
%d\n", pos, result);
```

```
    return 0;
```

```
}
```


#OUTPUT#

Output

Clear

Enter a number: 29

Enter bit position to keep (0-based index): 3

Result after clearing all bits except position 3 = 8

=== Code Execution Successful ===

9. Write a program to blink a led connected to pin number 2 of a microcontroller. Use software delay.

```
#code
#include <REGX51.H>

sbit LED = P2^0;

void delay() {
    unsigned int i, j;
    for(i = 0; i < 1000; i++) {
        for(j = 0; j < 100; j++);
    }
}

void main() {
    while(1) {
        LED = 0;
        delay();
        LED = 1;
        delay();
    }
}
```

#OUTPUT#

Output

LED connected to Port 2 Pin 0 (P2.0) blinks continuously.

State sequence:

LED ON

LED OFF

LED ON

LED OFF

...

10. Write an embedded C program to check the temperature. If it is more than 35 degree Centigrade, send high signal to fan so that it can turn on.

```
#code
```

```
#include <REGX51.H>
```

```
sbit FAN = P2^0;
```

```
void main()
```

```
{
```

```
    unsigned char temperature;
```

```
    while(1)
```

```
    {
```

```
        temperature = P1;
```

```
        if(temperature > 35)
```

```
        {
```

```
            FAN = 1;
```

```
        }
```

```
        else
```

```
        {
```

```
            FAN = 0;
```

```
        }
```

```
    }
```

```
}
```

#Output#

Temperature	Fan Status
30°C	OFF
36°C	ON
40°C	ON
25°C	OFF